

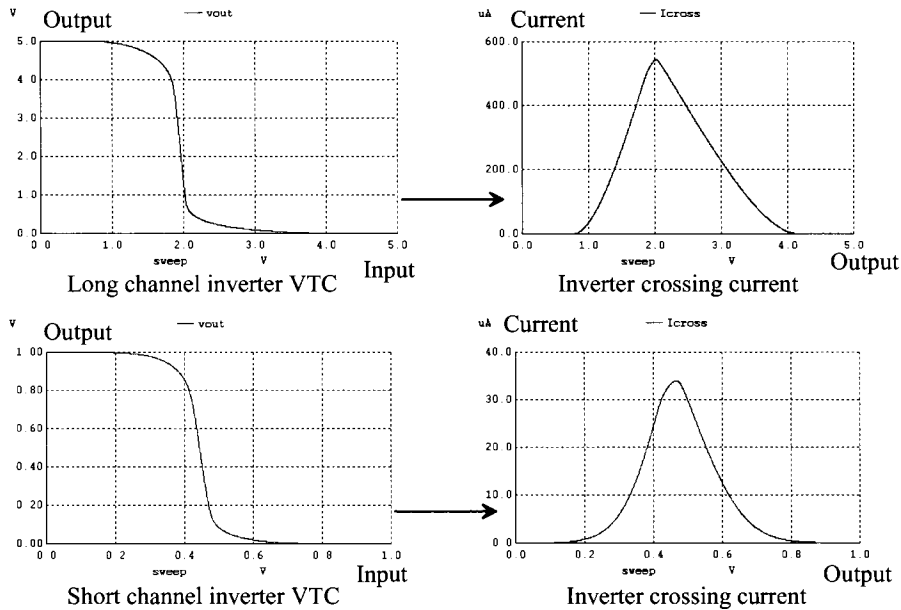


Figure 11.4 Voltage transfer curves (VTCs) for a long and a short channel inverter.

long-channel process, V_{IL} is approximately 1.8 V while, when using the short-channel process, V_{IL} is approximately 400 mV. For V_{IH} we get 2.1 V and 500 mV for the long- and short-channel processes, respectively.

Note that in Fig. 11.4 we also plotted the crossing current (the inverter's output voltage crossing between a logic 1 and a logic 0). This is the current that flows when the inverter is operating in region 2 in Fig. 11.2 (the inverter's input transitioning from a high to a low or from a low to a high). If the inverter's input transitions quickly, the amount of charge pulled from V_{DD} (the amount of time the inverter is operating in region 2) is small. However, if the inverter's input logic signal transitions slowly or the logic levels don't swing all the way to the power supply rails (like what we get with the pass gates discussed in the last chapter, see Fig. 10.19), it's possible for a significant current to flow through the inverter (important!) ■

Noise Margins

The noise margins of a digital gate or circuit indicate how well the gate will perform under noisy conditions. The noise margin for the high logic levels is given by

$$NM_H = V_{OH} - V_{IH} \quad (11.1)$$

and the noise margin for the low logic levels is given by

$$NM_L = V_{IL} - V_{OL} \quad (11.2)$$

For $V_{DD} = 1$ V, the ideal noise margins are 500 mV; that is, $NM_L = NM_H = V_{DD}/2$.

Inverter Switching Point

Consider the transfer characteristics of the basic inverter as shown in Fig. 11.5. Point C corresponds to the point on the curve when the input voltage is equal to the output voltage. At this point, the input (or output) voltage is called the *inverter switching point voltage*, V_{SP} , and both MOSFETs in the inverter are in the saturation region. Since the drain current in each MOSFET must be equal, the following is true:

$$\frac{\beta_n}{2}(V_{SP} - V_{THN})^2 = \frac{\beta_p}{2}(V_{DD} - V_{SP} - V_{THP})^2 \quad (11.3)$$

Solving for V_{SP} gives

$$V_{SP} = \frac{\sqrt{\frac{\beta_n}{\beta_p}} \cdot V_{THN} + (V_{DD} - V_{THP})}{1 + \sqrt{\frac{\beta_n}{\beta_p}}} \quad (11.4)$$

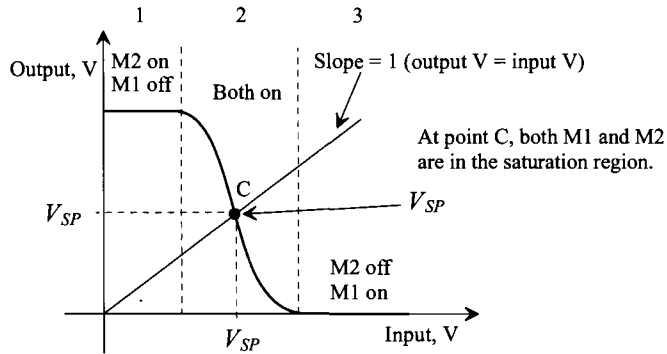


Figure 11.5 Transfer characteristics of the inverter showing the switching point.

Ideal Inverter VTC and Noise Margins

The ideal voltage transfer curves for an inverter are seen in Fig. 11.6. The ideal switching point voltage, V_{SP} , is $V_{DD}/2$. As seen in Eqs. (11.1) and (11.2), this makes the noise margins equal to ensure the best performance (a noise margin, say the logic low level, isn't improved at the cost of the other margin). When looking at Fig. 11.6, notice that, unlike what is seen in Fig. 11.5, the inverter never operates where both MOSFETs are on. The input to the inverter is recognized as either a 1 or a 0.

Example 11.2

Estimate β_n and β_p so that the switching point voltage of a CMOS inverter designed in the long-channel CMOS process is 2.5 V ($= V_{DD}/2$).

Solving Eq. (11.4) with $V_{SP} = 2.5$ V for the ratio β_n/β_p gives a value of approximately unity. That is,

$$\beta_n = \beta_p = KP_n \frac{W_1}{L_1} = KP_p \frac{W_2}{L_2}$$

Since $KP_n = 3KP_p$, the width of the PMOS device must be three times the width of the NMOS, assuming equal-length MOSFETs. For $V_{SP} = 2.5$ V, this requires

$$W_2 = 3W_1$$

which is also the requirement for making $R_n = R_p$. This is an important practical result. It shows how the electron and hole mobilities are related to both the effective switching resistances and the switching point voltage. As seen in Table 10.2, we often increase the widths of the PMOS devices to try to center V_{SP} and equate the propagation delays (the pull-up resistance, R_p , is the same as the pull-down resistance R_n). ■

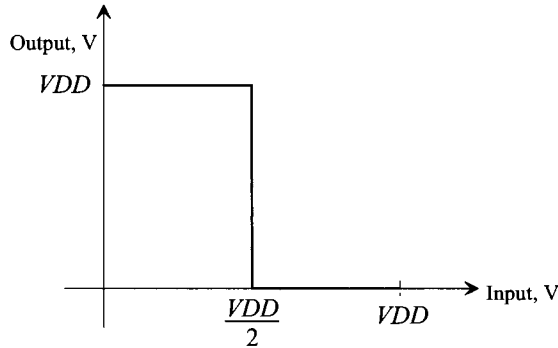


Figure 11.6 Ideal VTCs for the inverter.

Example 11.3

Show, using SPICE, that the inverter seen in Fig. 11.7 and implemented in the short-channel CMOS process has a switching point voltage close to the ideal value of $V_{DD}/2$. Comment on the sizes of the devices.

The simulation results are seen (also) in Fig. 11.7. Because we've sized the width of the PMOS to twice the width of the NMOS (see Table 10.2), the switching point is close to the ideal value of $V_{DD}/2$. We know that short-channel devices don't follow the square-law models and so we can't use Eq. [11.4] to calculate the V_{SP} in our 50 nm process. We can, however, get a good estimate using the effective switching resistances as seen in the figure (a voltage divider). When $R_p = R_n$, the switching point voltage is close to ideal. By changing the widths of the devices, we can adjust the switching point voltage. In other words, for a short-channel CMOS process we can use

$$V_{SP} = V_{DD} \cdot \frac{R_n}{R_n + R_p} \quad (11.5)$$

to estimate V_{SP} . This equation does have limitations. For example, if $R_n \gg R_p$, then this equation indicates V_{SP} is V_{DD} . However, we know that V_{SP} has to be

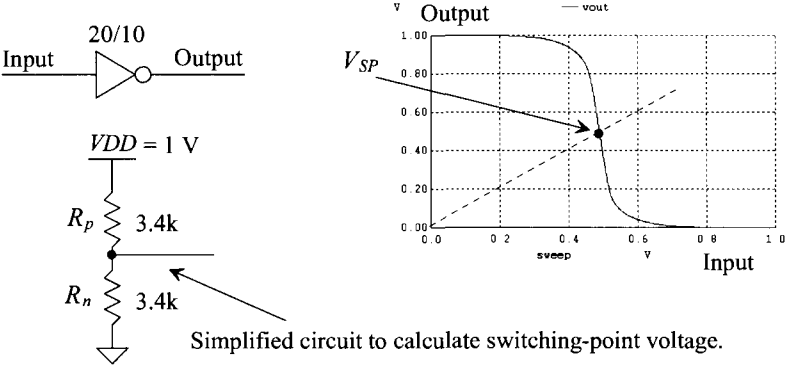


Figure 11.7 Switching point voltage for an inverter in the short-channel process.

greater than V_{THN} and less than $V_{DD} - V_{THP}$ (make sure that this is understood). ■

Example 11.4

Show, using SPICE and the long-channel process, the transfer curves for the CMOS inverter with transconductance ratios β_n/β_p of 3, 1, and $1/3$. Explain what changing the inverter ratio does to the transfer characteristics.

The simulation results are seen in Fig. 11.8. For all three DC sweeps the MOSFET's lengths are 1. For the case when $\beta_n/\beta_p = 1$, the $W_n = 10$ and $W_p = 30$. For the case when $\beta_n/\beta_p = 3$, the $W_n = 10$ and $W_p = 10$, etc. Increasing the strength of the NMOS (increasing the NMOS's width, which decreases R_n) causes the switching point voltage to decrease. We also get a decrease in V_{SP} by decreasing the strength of the PMOS device (which increases R_p). ■

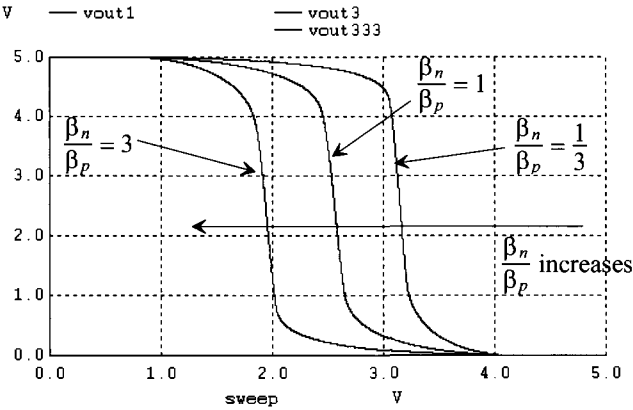


Figure 11.8 Sizing the inverter changes the switching point voltages.

11.2 Switching Characteristics

The switching behavior of the inverter can be generalized by examining the parasitic capacitances and resistances associated with the inverter. Consider the inverter shown in Fig. 11.9 with its equivalent digital model. Although the model is shown with both switches open, in practice one of the switches is closed, keeping the output connected to VDD or ground. The effective input capacitance of the inverter is

$$C_{in} = \frac{3}{2}(C_{ox1} + C_{ox2}) = C_{inn} + C_{inp} \quad (11.6)$$

The effective output capacitance of the inverter is simply

$$C_{out} = C_{ox1} + C_{ox2} = C_{outn} + C_{outp} \quad (11.7)$$

The intrinsic propagation delays of the inverter are

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{out} \text{ and } t_{PHL} = 0.7 \cdot R_{n1} \cdot C_{out} \quad (11.8)$$

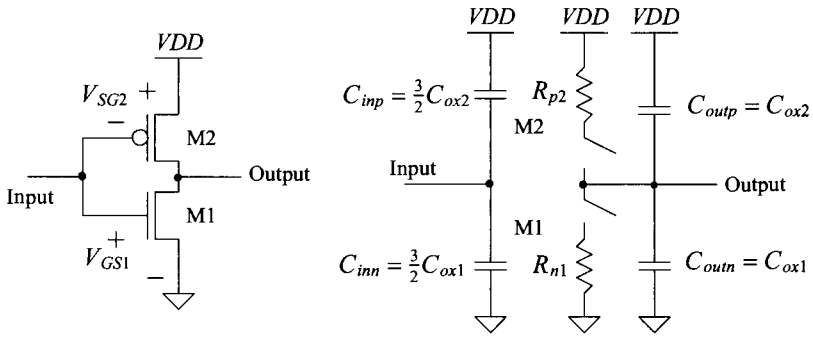


Figure 11.9 The CMOS inverter switching characteristics using the digital model.

Example 11.5

Estimate and simulate the intrinsic propagation delays of the inverter seen in Fig. 11.7. Estimate the inverter's input capacitance.

From the data in Table 10.2 and Eqs. (11.6) to (11.8), we can write

$$t_{PHL} = t_{PLH} = 0.7 \cdot 3.4k \cdot (0.625 + 1.25) \text{ fF} = 4.5 \text{ ps}$$

The simulation results are seen in Fig. 11.10. The intrinsic delays are considerably larger than this calculation (around 20 ps).

Using Eq. (11.6), the inverter's input capacitance is

$$C_{in} = \frac{3}{2}(0.625 + 1.25) \text{ fF} = 2.8 \text{ fF}$$

Sizing up the width of the PMOS so that its effective resistance is equal to R_n has the unwanted effect of increasing the inverter's input capacitance. ■

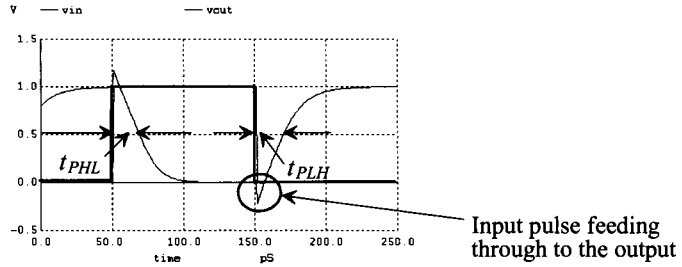


Figure 11.10 The intrinsic propagation delays of an inverter.

The propagation delays for an inverter driving a capacitive load are

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{tot} = 0.7 \cdot R_{p2} \cdot (C_{out} + C_{load}) \quad (11.9)$$

and

$$t_{PHL} = 0.7 \cdot R_{n1} \cdot C_{tot} = 0.7 \cdot R_{n1} \cdot (C_{out} + C_{load}) \quad (11.10)$$

where C_{tot} is the total capacitance on the output of the inverter, that is, the sum of the output capacitance of the inverter, any capacitance of interconnecting lines, and the input capacitance of the following gate(s).

Example 11.6

Estimate and simulate the propagation delays for the circuit seen in Fig. 11.11. Use the 50 nm CMOS process.

Because the load capacitance is much larger than the output capacitance of the inverter, we can rewrite

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{tot} \approx 0.7 \cdot R_{p2} \cdot C_{load} = 120 \text{ ps}$$

and

$$t_{PHL} = 0.7 \cdot R_{n1} \cdot C_{tot} \approx 0.7 \cdot R_{n1} \cdot C_{load} = 120 \text{ ps}$$

The simulation results are seen in Fig. 11.11. Notice that we are treating the MOSFETs used in the digital circuits as resistors and calculating the delays with a simple product of these resistors with the load capacitance. ■

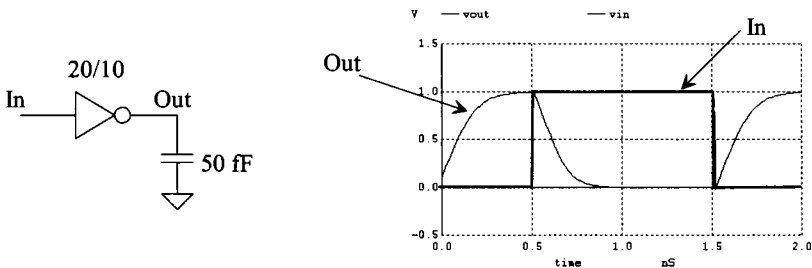


Figure 11.11 The delay associated with an inverter driving a 50 fF load.

The Ring Oscillator

The odd number of inverters in the circuit shown in Fig. 11.12 forms a closed loop with positive feedback and is called a ring oscillator. The oscillation frequency is given by

$$f_{osc} = \frac{1}{n \cdot (t_{PHL} + t_{PLH})} \quad (11.11)$$

assuming that the inverters are identical and n is the number (odd) of inverters in the ring oscillator. Since the ring oscillator is self-starting, it is often added to a test portion of a wafer to indicate the speed of a particular process run. The sum of the high-to-low and low-to-high delays is used to calculate the period of the oscillation because each inverter switches twice during a single oscillation period.

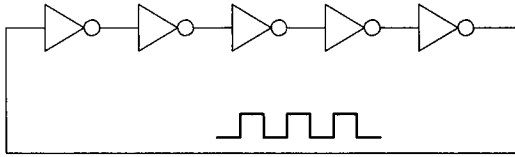


Figure 11.12 A five-stage ring oscillator.

When identical inverters are used the capacitance on the inverter's input/output is the sum of an inverter's input capacitance with the inverter's output capacitance, see Fig. 11.9, or

$$C_{tot} = \overbrace{C_{oxp} + C_{oxn}}^{C_{out}} + \overbrace{\frac{3}{2} \cdot (C_{oxp} + C_{oxn})}^{C_{in}} = \frac{5}{2} \cdot (C_{oxp} + C_{oxn}) \quad (11.12)$$

where, again, $C_{oxp} = C'_{ox} \cdot W_p \cdot L_p \cdot (scale)^2$ and $C_{oxn} = C'_{ox} \cdot W_n \cdot L_n \cdot (scale)^2$. The delay is then calculated using

$$t_{PHL} + t_{PLH} = 0.7 \cdot (R_n + R_p) \cdot C_{tot} \quad (11.13)$$

Dynamic Power Dissipation

Consider the CMOS inverter driving a capacitive load shown in Fig. 11.13. Each time the inverter changes states, it must either supply a charge to C_{tot} or sink the charge stored on C_{tot} to ground. If a square pulse is applied to the input of the inverter with a period T and frequency, f_{clk} , the average amount of current that the inverter must pull from VDD , recalling that current is being supplied from VDD only when the PMOS device is on, is

$$I_{avg} = \frac{Q_{C_{tot}}}{T} = \frac{VDD \cdot C_{tot}}{T} \quad (11.14)$$

The average dynamic power dissipated by the inverter is

$$P_{avg} = VDD \cdot I_{avg} = \frac{C_{tot} \cdot VDD^2}{T} = C_{tot} \cdot VDD^2 \cdot f_{clk} \quad (11.15)$$

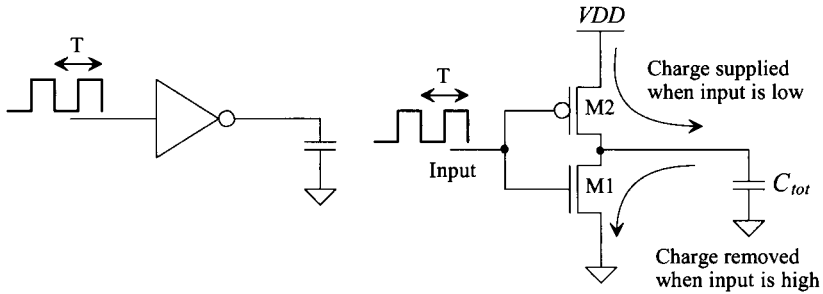


Figure 11.13 Dynamic power dissipation of the CMOS inverter.

Notice that the power dissipation is a function of the clock frequency, power supply voltage, and load capacitance. A great deal of effort is put into reducing the power dissipation in CMOS circuits. One of the major advantages of dynamic logic (Ch. 14) is its lower power dissipation.

To characterize the speed of a digital process, a term called the power delay product (*PDP*) is often used. The *PDP*, measured in Joules, is defined by

$$PDP = P_{avg} \cdot (t_{PHL} + t_{PLH}) \quad (11.16)$$

These terms can be determined from a ring oscillator. The *PDP* is frequently used to compare different technologies or device sizes; for example, a GaAs process can be compared with a 50 nm CMOS process. Although the GaAs process may have a lower propagation delay, the power dissipation may be larger and result in a larger *PDP*.

Example 11.7

Estimate the oscillation frequency of an 11-stage ring oscillator in the 50 nm process using the inverter seen in Fig. 11.7 (see Table 10.2). Verify the estimate with simulations.

Using the data in Table 10.2 and Eqs. (11.11) – (11.13), we can write

$$C_{tot} = \frac{5}{2} \cdot (1.25 + 0.625) \text{ fF} = 4.7 \text{ fF}$$

and

$$t_{PHL} + t_{PLH} = 0.7 \cdot (3.4k + 3.4k) \cdot 4.7 \text{ fF} = 22 \text{ ps}$$

For an 11-stage ring oscillator, we get an oscillation frequency of

$$f_{osc} = \frac{1}{11 \cdot (22 \text{ ps})} = 4.1 \text{ GHz}$$

The simulation results are seen in Fig. 11.14. The simulated oscillation frequency is close to 1.25 GHz or considerably different from our hand calculations (as will be the case when delays are close to the intrinsic values). The average power, P_{avg} , dissipated by a single inverter in this ring oscillator, using Eq. (11.15), is estimated as 19.6 μW . The *PDP* for the 50 nm process is then $431 \times 10^{-18} \text{ J}$. ■